

Functional Requirement Document / Technical Specification Document

Project Name: **InsureDataPipelineData**: October, 2025

Revision Sheet

Release No.	Date	Revision Description	Owner
1	13-Oct-2025	First Version of InsureDataPipelineDataData Intelligence: Sept, 2025	IntelliBI

1.0 PURPOSE

The purpose of this document is to define the functional and technical requirements for building a **InsureDataPipeline (IDP)** using **Python, Pandas, File Automations** aimed at Insurance data from multiple source files into a centralized persistent system.

Using Python file processing and automation will be handled.

The primary objectives include:

- Automate file flow, Validation and archival of files.
- Applying **business logic** to maintain **data quality** and consistency across various datasets.
- Enabling **real-time Analytics and** through Python Script.

2.0 SCOPE

The scope of this project includes:

- **File Validation** from source systems like flat files (.csv)
- **Data Cleaning** and load it into a preprocessed Layer.
- **Load the data** into a semantic layer for tracking aggregate data.

3.0 ASSUMPTIONS AND CONSTRAINTS

Assumptions:

- **Business Rules:** Clear business rules for data validation are defined.
- **Data Quality:** Source systems provide accurate data with minimal errors.
- **Tool Availability:** (Python, Pandas etc.) are pre-configured.
- **Stakeholder Engagement:** Business stakeholders are available for input.

Constraints:

1. **Data Volume:** High volumes of data may strain Python resources.

2. **Real-time Reporting:** Near real-time, not true real-time reporting due to processing delays.
3. **Data Source Delays:** Pipeline depends on timely data from external sources.

4.0 DESIGN CONSIDERATIONS

4.1 Naming Standards

FileName:

Convention: **lower_case** (all lowercase, words separated by underscores)

Example : data_ingestion.py

FunctionName:

Convention: **lower_case** (all lowercase, words separated by underscores)

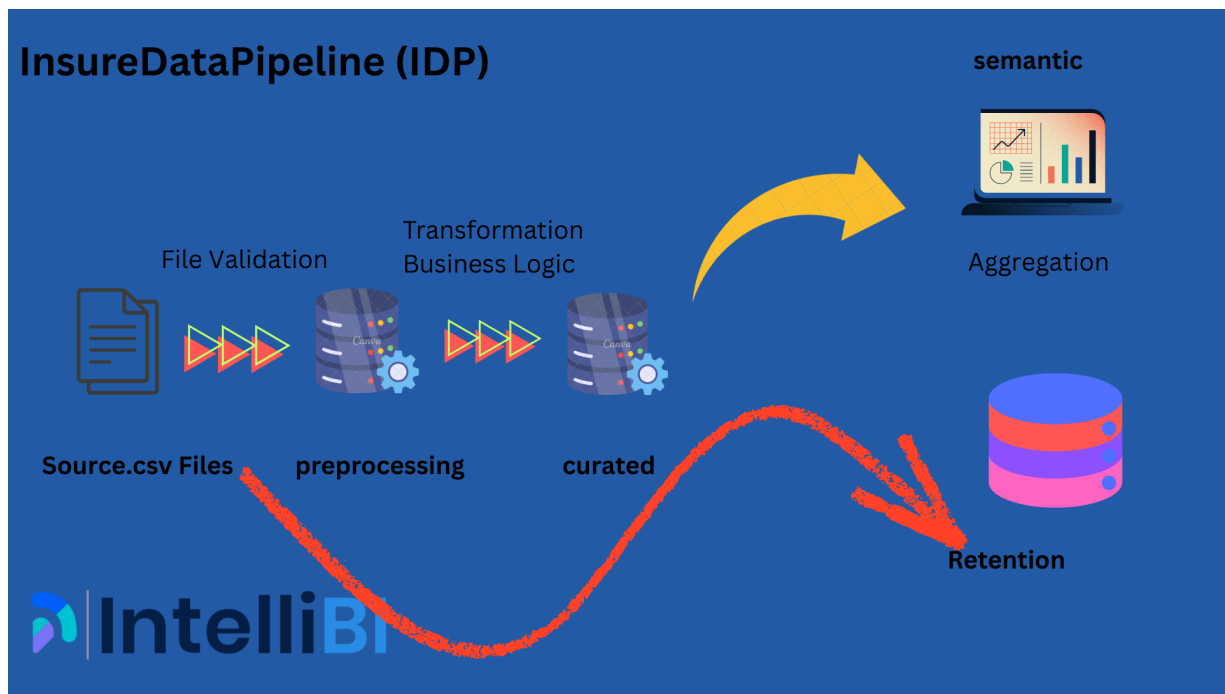
Example : def transform_claim_data():

FolderName:

Convention: **lower_case** (all lowercase, words separated by underscores)

Example : semantic_layer

5.0 HIGH LEVEL DESIGN (PROJECT ARCHITECTURE DIAGRAM)



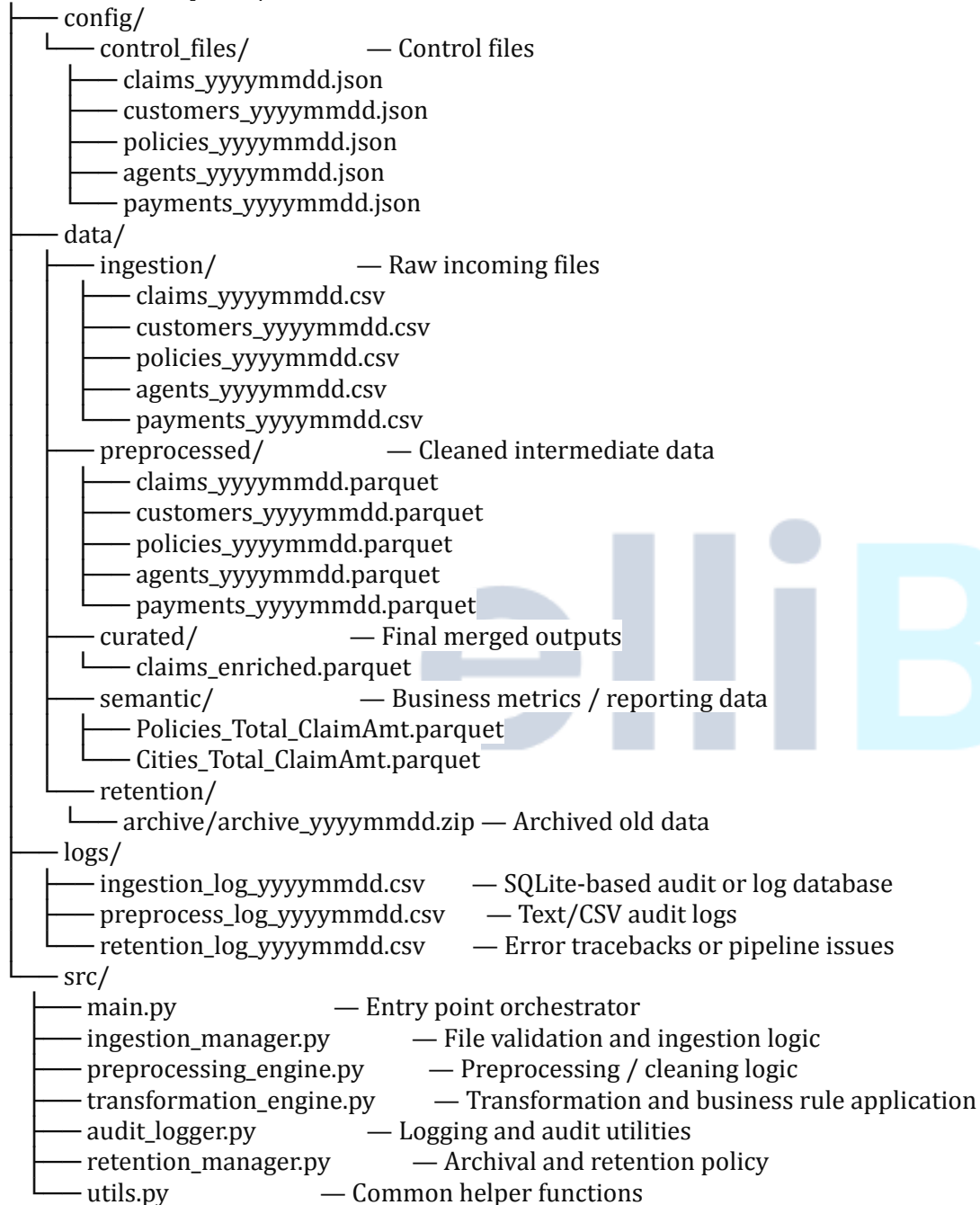
Technology components

Technology	Function
Python	For File Validation, Archival
Ad.Python	Build PipeLine and Process the data.

ENVIRONMENTS SETUP: DEVELOPMENT

Create Folder Structure as follows for **dev** Environment:

devInsureDataPipeline/



Create Folder Structure as follows for **QA** Environment:

1.

Create Folder Structure as follows for **Prod** Environment:

2.

Note: Set up only in the **DEV** environment for now.

1. Inbound Layer:

1. Inbound Layer Requirements

1. File Validation Control Files

- o **devInsureDataPipeline/config/control_files/**
 - claims_yyyymmdd.json
 - customers_yyyymmdd.json
 - policies_yyyymmdd.json
 - agents_yyyymmdd.json
 - payments_yyyymmdd.json

2. File Validation Data Files

- o
 - claims_yyyymmdd.csv
 - customers_yyyymmdd.csv
 - policies_yyyymmdd.csv
 - agents_yyyymmdd.csv
 - payments_yyyymmdd.csv

3. Files to Fetch from Source:

Source Files Location: [Source Files](#)

Control Files Location: [Control Files](#)

Files to be Validated based on Control file and Data file:

- If File is valid then load the data into the preprocessed layer.

4. Preprocessed Layer Data Load:

Flat File :

Load the data from the **Source Files** to the **Preprocessed** layer with some basic data cleaning and business logic and audit Columns.

1. Remove duplicate values .
2. Add **ingestion_date** column and load system date into **load_date** column.
3. Add **file_date** column and load file date into it.

Load Type: Truncate & Load

Unit Test Case Documents: [Unit Test Case Document Template](#)

Data Model: [Preprocessed Layer Data Model](#)

STTM : [STTM Source to Preprocessed Layer](#)

5. **Preprocessed to Curated Data Load:**

Load the data from the RAW layer (**Preprocessed**) to the (**Curated**) layer with some basic business logic Columns.

Load Type: Incremental Load

STTM Preprocessed to Curated: [STTM_Preprocessed to Curated](#)
Data Model: [Cleansed Layer Data Model](#)

6. **Curated to Semantic Layer Tables:**

Load Type: Incremental Load

Please refer to the attached Document for this.

Unit Test Case Documents: [STTM Curated to Semantic](#)
Data Model: [Refined Layer Data Model](#)

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Orchestration For Flat File Source:

1. File Validation & Source Files To Preprocessed:

- 1. Create end-to-end pipeline (ingestion_manager.py) for each .csv File :**
 - o Create function **validate_file()**
 - o Validate the files and process only valid files.
- 2. Create end-to-end pipeline (preprocessing_engine.py) :**
 - o Create function **preprocess_file()**
 - o Load the data into parquet files with audit columns.
- 3. Create end-to-end pipeline (transformation_engine.py) :**
 - o Create function **build_curated_and_semantic()**
 - o Load the data into parquet files with required business logic.
- 4. Create end-to-end pipeline (retention_manager.py) :**
 - o Create function **archive_file()**
 - o Move the processed files into the retention/archive folder.
- 5. Create end-to-end pipeline (audit_logger.py) :**
 - o Create function **archilog_ingestion(), log_preprocess(), log_retention()**.
 - o Generate the log Files
Log Files Format:
- 6. Create end-to-end Master pipeline (main.py) :**
 - o Execute the below functions from **run_pipeline()**
 - o **run_pipeline()**,
 - **validated_files()**
 - **preprocess_file()**
 - **build_curated_and_semantic()**
 - **archive_file()**

4. Scheduling:

- Schedule the master pipeline ([main.py](#)) to run daily at 10 PM IST.